

Reading Notes on the Development of Large Language Models

Yiding Sun

1 Attention Is All You Need

The main contribution of *Attention Is All You Need* is the Transformer architecture [1]. Before the Transformer, many sequence models relied on recurrent neural networks, especially RNNs and LSTMs. These models processed tokens one after another, which made it difficult to parallelize the computation. The Transformer removed recurrence and instead used attention as its main mechanism.

The central idea of self-attention is that each token can examine the other tokens in the sequence and decide which ones are important. The model creates query, key, and value vectors for each token. The similarity between a query and a key determines how much attention should be given to the corresponding value. Multi-head attention repeats this process in several representation spaces, allowing different attention heads to capture different relationships.

One detail I found important is positional encoding. Since the Transformer does not process words sequentially, it does not automatically know their order. Positional information must therefore be added to the input embeddings. This showed me that removing recurrence solves one problem but also creates another problem that must be handled explicitly.

My main impression is that the Transformer changed the basic way of thinking about language models. Instead of forcing information to pass through a sequence step by step, it allows tokens to interact more directly. I do not yet understand every matrix operation in the model, but I can see why this architecture is faster to train and better at connecting words that are far apart.

2 BERT: Pre-training of Deep Bidirectional Transformers

BERT applies the Transformer architecture to language understanding through bidirectional pre-training [2]. Unlike an autoregressive model that predicts text only from the previous words, BERT attempts to build a representation using both the left and right context.

Its main training task is Masked Language Modeling, or MLM. Some tokens in a sentence are replaced by a special [MASK] token, and the model learns to reconstruct them. Because the hidden word may depend on words on both sides, the model is encouraged to learn bidirectional contextual representations. The original paper also uses Next Sentence Prediction to train the model to identify whether one sentence follows another.

Another important idea is the combination of pre-training and fine-tuning. BERT is first trained on a large amount of unlabeled text. It can then be adapted to tasks such as question answering, sentiment classification, and natural language inference with relatively small modifications. This is more efficient than designing and training a completely separate model for every task.

What interested me most is that a word does not have only one fixed representation. Its representation depends on the sentence in which it appears. For example, the word “bank” should be represented differently in “river bank” and “bank account.” This idea seems simple from a human perspective, but BERT provides a systematic way for a neural network to learn it.

I also noticed that BERT is more focused on understanding existing text than freely generating long passages. This helped me understand why different Transformer-based models can still have very different purposes.

3 Language Models Are Few-Shot Learners

This paper introduces GPT-3, an autoregressive language model with 175 billion parameters [3]. Its main finding is that sufficiently large language models can perform many tasks from instructions and a few examples, without updating their parameters for each new task.

The paper compares zero-shot, one-shot, and few-shot learning. In zero-shot learning, the model receives only a task instruction. In one-shot learning, it receives one example, while few-shot learning provides several examples in the prompt. The model then continues the text in a way that follows the demonstrated pattern. This process is now often called in-context learning.

I found this idea surprising because the model is not being trained again in the usual sense. Its parameters remain unchanged, but it can temporarily adapt its behavior based on the context. The prompt therefore becomes part of the method rather than only a question given after training.

At the same time, the paper does not show that scaling solves every problem. GPT-3 still performs poorly on some reasoning tasks, and its results can depend heavily on how the prompt is written. Training on large collections of internet text also introduces concerns about bias, misinformation, and accidental overlap between training data and evaluation data.

My main takeaway is that scale can create abilities that are not obvious from smaller models. However, I do not think the paper proves that increasing model size is enough for general intelligence. It shows a strong empirical trend, but the source of the model’s abilities and limitations still needs further study.

4 Training Language Models to Follow Instructions with Human Feedback

The InstructGPT paper studies a problem that is different from simply predicting the next word: how to make a language model follow what users actually want [4]. A model trained only on internet text may produce fluent sentences without being helpful, truthful, or safe.

The training process has several stages. First, human labelers write examples of good responses, and the base GPT-3 model is fine-tuned on these demonstrations through supervised learning. Next, the model generates several answers to the same prompt, and human labelers rank these answers. These preferences are used to train a reward model. Finally, reinforcement learning is used to optimize the language model according to the learned reward. This overall approach is known as Reinforcement Learning from Human Feedback, or RLHF.

One result I found especially interesting is that the 1.3-billion-parameter InstructGPT model was often preferred by human evaluators over the original 175-billion-parameter GPT-3. This suggests that model size and usefulness are not the same thing. A smaller model can give better answers when its training objective is more closely connected to human expectations.

However, human feedback also raises questions. The reward model learns from the judgments of a limited group of labelers, and their preferences may not represent every user or culture. In addition, a model can produce an answer that sounds helpful even when the information is incorrect. Therefore, alignment through human feedback seems useful, but it is not a complete solution to reliability.

5 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

This paper introduces chain-of-thought prompting, in which the model is shown intermediate reasoning steps before producing a final answer [5]. Instead of giving only question-answer examples, the prompt demonstrates how a problem can be solved step by step.

The method is simple compared with changing the model architecture or training process. For example, in a mathematical word problem, the demonstration explains which quantities are relevant, how they are combined, and why the final answer follows. The language model then attempts to produce a similar reasoning process for a new problem.

The experiments show that chain-of-thought prompting can improve performance on arithmetic, commonsense, and symbolic reasoning tasks, especially for very large models. This is connected to few-shot learning because the reasoning pattern is provided through examples in the prompt rather than through additional parameter updates.

As a mathematics student, I found this paper easy to relate to. In mathematics classes, writing only the final answer is usually not enough; the intermediate derivation shows whether the reasoning is valid. Chain-of-thought prompting gives language models a similar working space.

Still, I think there is an important limitation. A written reasoning chain may look logical even when one of its steps is false. Therefore, generating more steps does not automatically guarantee correct reasoning. The method improves performance, but the displayed explanation should not always be treated as a completely faithful description of what happened inside the model.

6 Overall Reflection

Together, these papers gave me a clearer picture of how modern large language models developed. The Transformer provided an efficient architecture based on attention. BERT showed how bidirectional pre-training could create useful language representations. GPT-3 demonstrated the importance of scale and in-context learning. InstructGPT used human feedback to improve instruction following, while chain-of-thought prompting showed that the format of a prompt could affect reasoning performance.

A simplified way to describe this development is that researchers first improved how models process language, then expanded what they could learn from large datasets, and finally tried to make their behavior more useful and understandable to humans. These stages are connected, but each solves a different problem.

After reading the papers, I am more impressed by the interaction between mathematical ideas and practical design choices. At the same time, I became more cautious about words such as “understanding” and “reasoning.” Strong experimental performance does not completely explain how a model reaches its answer. For me, this is one of the most interesting open questions in the study of large language models.

References

- [1] A. Vaswani et al., “Attention Is All You Need,” *Advances in Neural Information Processing Systems*, 2017. Available at: <https://arxiv.org/abs/1706.03762>.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of NAACL-HLT*, 2019. Available at: <https://arxiv.org/abs/1810.04805>.
- [3] T. Brown et al., “Language Models Are Few-Shot Learners,” *Advances in Neural Information Processing Systems*, 2020. Available at: <https://arxiv.org/abs/2005.14165>.
- [4] L. Ouyang et al., “Training Language Models to Follow Instructions with Human Feedback,” *Advances in Neural Information Processing Systems*, 2022. Available at: <https://arxiv.org/abs/2203.02155>.
- [5] J. Wei et al., “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” *Advances in Neural Information Processing Systems*, 2022. Available at: <https://arxiv.org/abs/2201.11903>.